

Heyting Algebras in Dedekind-Macneille Completed Causal Structure of Minkowski Space: a No-Go Theorem

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May 2026

Abstract

The Minkowski metric induces a causal structure on events in Minkowski space. We complete the causal structure of spacetime events with a Dedekind-Macneille completion, thus inducing a complete lattice structure of causal pasts. Via counterexample, we prove this lattice is not infinity distributive and therefore not a Heyting algebra. This implies that the Dedekind completion of causal structure, as an internal logic, is not sufficiently powerful to guarantee the Heyting algebra implication.

1 Background

We will first define Minkowski space and causal structure. Then, using tools from lattice theory, we construct causal pasts and their ‘ideal’ overlaps. This construction induces an internal model of logic of spacetime. This is close in spirit to analyzing the topological structures induced by Minkowski causal structure [1]. A lattice can be viewed as a weaker version of a topological space in which the arbitrary union axiom is dropped; these are called *closure systems* and are a focal point in *pointless topology*.

1.1 Spacetime Definitions

To study the causality of space-time events in special relativity, we define a manifold called *Minkowski Space* M with the Minkowski metric $\eta(X, Y)$ as follows:

Definition 1.1 (Minkowski Metric Tensor). The Lorentzian metric tensor for space-time is given by:

$$\eta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad (1)$$

Definition 1.2 (Minkowski Space). We define a manifold M with a Minkowski metric η as Minkowski Space when M has zero curvature.

Remark 1.1. Our physical model uses $M \subseteq \mathbb{R}^4$

Definition 1.3 (Minkowski Metric). Given two vectors $X, Y \in \mathbb{R}^4$, the *Minkowski metric* $g(X, Y)$ is given by $X^T \eta Y$

If we take a smooth, regular parametric curve $\gamma \subset M$, this describes the trajectory of a particle in space-time. If we sample a point X on this trajectory, this gives us a space-time event. The tangent vectors of the events of the parametric curve can be classified in three ways. Doing so allows us to describe a relationship between the causality of events with specific classes of parametric curves.

Definition 1.4 (Event). Given a regular smooth curve $\gamma \subset M$, we call $X \in \gamma$ an *event*.

Remark 1.2. The tangent space $T_X M$ is the vector space of all tangent vectors to the point $X \in \gamma$.

Definition 1.5 (Lightlike Events). Given a vector $Y \in T_X M$, Y is called *lightlike* if $g(Y, Y) = 0$.

Definition 1.6 (Timelike Events). Given a vector $Y \in T_X M$, Y is called *timelike* if $g(Y, Y) > 0$.

Definition 1.7 (Spacelike Events). Given a vector $Y \in T_X M$, Y is called *spacelike* if $g(Y, Y) < 0$.

Remark 1.3. If a particle is moving with the velocity $v = c$ at every point on its trajectory, this particle's trajectory is lightlike. If the particle is moving with velocity $v < c$, then this particle's velocity is timelike. $v > c$ is forbidden by Einstein's axioms of special relativity.

Definition 1.8 (Timelike Curve). A *timelike curve* is a smooth regular parametric curve $\gamma \subset M$, $\gamma(u) : [u_0, u_1] \rightarrow \mathbb{R}^4 :: u \mapsto (ct(u), x(u), y(u), z(u))$, $\gamma(t_0) = X$, $\gamma(t_1) = Y$ such that every tangent vector is timelike. That is, for all $u \in \mathbb{R}$, $g(\dot{\gamma}(u), \dot{\gamma}(u)) > 0$ is timelike.[2]

We will not go into the details of time-orientability, but we should mention that we assume time-orientability in the sense that for every event, that event's light cone can be assigned a future and a past. Future-directed curves are then time-like curves whose tangent vectors live in the future light-cones.

Our classification of events induces causal structure as follows:

Definition 1.9 (Causal Ordering). Given events $X, Y \in M$, X *causally precedes* Y iff there exists a future-directed timelike curve $\gamma \subset M$, $\gamma : [u_0, u_1] \rightarrow \mathbb{R}^4$, $\gamma(u_0) = X$, $\gamma(u_1) = Y$ or if $X = Y$. This is written $X \prec Y$.

We define causal future and causal past of events in-terms of their *upper-sets* and *lower-sets* respectively.

Definition 1.10 (Causal Future). The *causal future* of $x \in M$ is defined as $I^+(x) := \{y \in M \mid x \prec y\}$.

Definition 1.11 (Causal Past). The *causal past* of $x \in M$ is defined as $I^-(x) := \{y \in M \mid y \prec x\}$.

Causal order is a *partial order* (see Def. 1.12).

Theorem 1.1. *Causal ordering on Minkowski space is a partial order.*

Proof. We must show transitivity, reflexivity, and antisymmetry of $\prec$. Take events $X, Y, Z \in M$.

1. **Reflexivity:** $X \prec X$ follows from $X = X$.
2. **Transitivity:** Assume $X \prec Y$ and $Y \prec Z$. Then there exist parametric curves $\alpha(u) : [0, 1] \rightarrow \mathbb{R}^4$ and $\beta(u) : [1, 2] \rightarrow \mathbb{R}^4$ from X to Y and Y to Z , respectively. Take the parametric curve $\gamma = \begin{cases} \alpha(u) & \text{if } 0 \leq u \leq 1 \\ \beta(u) & \text{if } 1 \leq u \leq 2 \end{cases}$. Then this new curve has all timelike tangent vectors and $\gamma(0) = X$, $\gamma(2) = Z$. Thus $X \prec Z$ ¹.
3. **Antisymmetry:** Take $X \prec Y$, $Y \prec X$ and suppose for the sake of contradiction that $X \neq Y$. This implies there exist timelike, future-directed curves $\gamma_1 : [0, 1] \rightarrow \mathbb{R}^4 :: X \mapsto Y$ and $\gamma_2 : [0, 1] \rightarrow \mathbb{R}^4 :: Y \mapsto X$. Thus, $ct_1 > 0$ and $ct_2 > 0$ for all $u \in [0, 1]$. Integrating both curves and applying the fundamental theorem of calculus, we obtain:

$$t(Y) - t(X) = \int_0^1 \frac{dt}{du} du > 0 \implies t(Y) > t(X) \quad (2)$$

and

$$t(X) - t(Y) = \int_0^1 \frac{dt}{du} du > 0 \implies t(X) > t(Y) \quad (3)$$

Thus $t(X) < t(Y) < t(X) \implies t(X) < t(X)$. This is a contradiction. Therefore, $X = Y$.

□

We have shown that M has a partial order. We call $(M, \prec)$ a *poset*². An interesting question to ask is whether $(M, \prec)$ forms a *lattice*. If we want $(M, \prec)$ to form a lattice, then for every $x, y \in M$, the set $\{x, y\}$ must have a unique *supremum* and *infimum*. Lattices with special properties can be thought of as a particular logical structure. Some examples are Boolean and Heyting algebras. Thus, by describing spacetime in terms of lattices with certain properties, we can understand the structure of spacetime causality as having induced an internal logic.

¹Technically this curve is only piecewise smooth.

²Some texts also refer to this as *quasi-order*

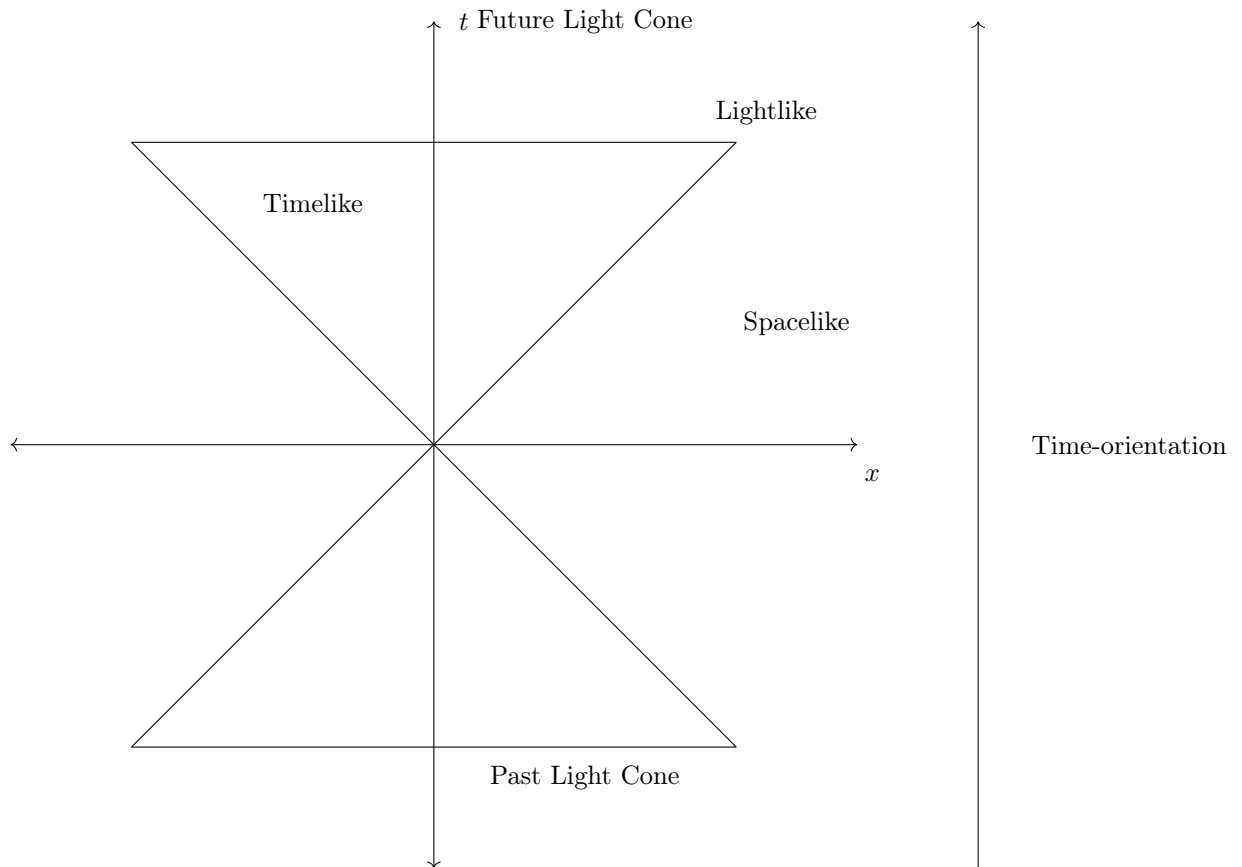


Figure 1: Lightcone with time-orientation and classifications of regions in space. This models a projection of a 4D tangent plane of an event in Minkowski space onto a 2D space. Notice that the light-like boundaries look like they're at an incline angle of $\pi/4$. This would be if we set $c = 1$.

1.2 Lattice Theory

We will briefly review lattice theory. Lattice theory is the mathematics of ordering. Simply using ordered sets, we can construct models of classical and intuitionistic logic, or describe the categorical nature of mathematical structures.

Definition 1.12 (Partial order). A *partial order* is a binary relation $\leq$ on a set S that is transitive, reflexive, and antisymmetric:

1. **Reflexive:** For every $x \in S$, $x \leq x$.
2. **Transitive:** If $x \leq y$, $y \leq z$, then $x \leq z$.

3. **Antisymmetric:** If $x \leq y$ and $y \leq x$, then $x = y$.

The underlying set S is called a *poset*, and we write the poset with its ordering as the 2-tuple $(S, \leq)$.

Definition 1.13 (Upperbound). Given a partial order $(S, \leq)$ and a subset $A \subseteq S$, x is an *upper bound* of A if for all $a \in A$, $a \leq x$.

Lower bounds are dually defined.

Definition 1.14 (Supremum). Given a partial order $(S, \leq)$, and a subset $A \subseteq S$, if for all upper bounds m , the upper bound $x \leq m$, then x is a *supremum*.

Remark 1.4. The *infimum* is defined dually to the supremum. When we say two definitions are *dual*, this means that if we invert the partial order of a set for one definition, the two definitions are the same. That is, for every the dual partial order is given by $x \leq y$ iff $y \leq x$ in the original⁴.

Definition 1.15 (Lattice). Given a partial order $(S, \leq)$, S is a lattice if for every pair of elements $x, y \in S$, there exist both a unique supremum and infimum of $\{x, y\}$. We denote the supremum as $x \vee y$ and infimum as $x \wedge y$.

We now define special properties of a lattice:

Definition 1.16 (Bounded). A lattice L is bounded if there exists a supremum $\top$ (called top) and infimum $\perp$ (called bottom) on the entire set L .

Definition 1.17 (Complements). Take a bounded lattice L . Then a pair $\{x, y\} \subseteq L$ is *complemented* if $x \vee y = \top$ and $x \wedge y = \perp$. We call x the *complement* to y and vice versa.

Definition 1.18 (Complemented Lattice). A bounded lattice L is *complemented* if every element $x \in L$ has a complement

Definition 1.19 (Heyting Algebra). A bounded lattice $(L, \leq)$ is called a *Heyting Algebra* if for every $x, y \in L$, there exists an element $(x \implies y) \in L$ such that, given an element $a \in L$, $a \leq (x \implies y) \iff a \wedge x \leq y$.

Definition 1.20 (Pseudo Complement). Given a Heyting algebra $(L, \leq)$ and $a \in L$, the *pseudo-complement* of a , denoted $\neg a$, is defined as $\neg a := a \implies \perp$.

Remark 1.5. A Heyting Algebra is a weaker structure than a *Boolean Algebra*. That is, every Boolean Algebra is also a Heyting Algebra. Boolean Algebras come with the additional rule that for every A in the Boolean algebra, $A \vee \neg A = \top$.

Heyting algebras give the weakest structure necessary to produce the rule of *modus ponens*:

$$\frac{P \quad P \implies Q}{Q}$$

³When we discuss subsets of our posets, we will automatically assume that these subsets have inherited the partial order.

⁴Supremum and infimum are also referred to as *joins* and *meets* respectively

where this is read as ‘P and P implies Q derives Q’.

Definition 1.21 (Completion). A *complete lattice* L is a lattice in which every subset has a unique infimum and supremum.

Remark 1.6. The distinction between a complete lattice and a lattice that is not necessarily complete is that, by induction, any lattice can have a unique supremum or infimum on its finite subsets, whereas a complete lattice has suprema and infima for arbitrarily large subsets.

Theorem 1.2 (Infinite Distributivity of Heyting Algebras). *All complete Heyting Algebras $(L, \leq)$ satisfy infinite distributivity in the sense that $A \wedge (\bigvee_{i \in S} B_i) = \bigvee_{i \in S} (A \wedge B_i)$, where S is a set indexing B_i .*

Proof. See [3]. □

1.3 Dedekind-Macneille Completion

Partial orders induced by the Minkowski metric do not necessarily satisfy being a lattice. $I^+(x) \cap I^+(y)$ gives the set of suprema of $\{x, y\}$, whilst $I^-(x) \cap I^-(y)$ give the infima of $\{x, y\}$. Notice that these intersections are entire hypersurface.

Theorem 1.3. $(M, \prec)$ does not satisfy the conditions to be a lattice.

Proof. We will show this via contradiction. Take events $A = (t_a, x_a)$, $B = (t_b, x_b)$. Suppose there exists a supremum $m = A \vee B = (t_m, x_m)$. Let $\epsilon > 0$. If m has a distance $m \cdot \hat{x} = \epsilon$ from the light-like boundary of the common future of A and B , then let $\delta = \epsilon/2$. Then take $\tilde{m} = (t_m, x_m + \delta)$. $\tilde{m}$ is also timelike and $\tilde{m}$ must also be an upper bound of A and B . There is no causal ordering between m and $\tilde{m}$, so $m \not\prec \tilde{m}$, which contradicts the definition of supremum. □

In order to recover a lattice, we take the Dedekind-Macneille (DM) completion of $(M, \prec)$, which gives us a lattice $(L, \subseteq)$.

Definition 1.22 (Dedekind-Macneille Completion). Given a partial order $(S, \leq)$ and a subset $A \subseteq S$, we define the set A^u as the set of upper bounds of A , and A^l as the set of lower bounds of A . Then we take all sets A of the form $(A^u)^l = A$. The completion of $(S, \leq)$ is given by the set L of all such A [4]. This new construction is called the Dedekind-Macneille (DM) Completion, and we denote the completed set as $DM(M)$. Trivially, if every subset has a unique supremum and infimum, then the lattice condition is satisfied and the partially ordered lattice is $(DM, \subseteq)$ ⁵.

Lemma 1.1. *Let the downset $\downarrow x := \{y \in S \mid y \leq x\}$. Then $\downarrow x$ is in the DM Completion.*

⁵For example, $DM(\mathbb{N}) = \mathbb{N} \cup \{\top\}$

Proof. We define $\uparrow x = \{y \in S \mid x \leq y\}$. Then $(\downarrow x)^u = \uparrow x \implies ((\downarrow x)^u)^l = \downarrow x$. $\square$

We can observe that the mapping $\phi : S \hookrightarrow DM(S) :: x \mapsto \downarrow x$ is an embedding isomorphism in the sense that $x \leq y \implies \phi(x) = \downarrow x \subseteq \downarrow y = \phi(y)$. This forms a completion [4].

We weren't able to argue about the logic of spacetime simply using events, but we can argue about the same structure using the causal pasts of events and their idealized causal pasts.

Remark 1.7. Given $X \in M$, the downset $\downarrow X$ physically corresponds to the causal past of an event X . So $\downarrow X \equiv I^-(X)$.

Remark 1.8. To find an idealized causal past of two events, one can take the intersection $I^-(X) \cap I^-(Y)$.

Cuts in this completion warrant a careful physical interpretation. If we take the supremum of two events $X, Y \in M$ as $X \vee Y$, this supremum typically results in a hypersurface. However, the introduction of a cut gives a unique supremum to both the causal pasts $\downarrow X$ and $\downarrow Y$. We call this an *idealized causal past*.

Remark 1.9. Because the Dedekind-Macneille completion of Minkowski causal structure is self-dual, this means that the logic induced by causal structure is invariant with respect to time-orientability.

Definition 1.23 (Idealized-Minkowski Lattice). Let $(DM(M), \subseteq)$ be the DM completion of the Minkowski space partial order $(M, <)$. We call $(DM(M), \subseteq)$ an *Idealized-Minkowski (IM) Lattice*.

Remark 1.10. The top element $\top$ corresponds to the causal past of the entire Minkowski space, whereas the bottom element $\perp$ corresponds to a causal past with no spacetime history. In terms of the propositional logic induced by Minkowski causal structure, $\top$ acts as a Boolean true, and $\perp$ acts as a boolean false. Thus, $\neg X$ for some causal past X is saying that the causal past X being true implies no causal history. More generally, modus ponens, as implicitly described in definition 1.19, says that if we have a causal past A that entails $X \implies Y$ and we also have a causal past X , then causal past Y must also be true. Thus, if the IM Lattice is a Heyting Algebra, the internal logic allows for deductive reasoning of causal histories. We will that the IM Lattice is not, in fact, a Heyting algebra. There are examples where we cannot deduce causal structures. Intersection, acting as our infimum $\wedge$, makes statements about common causal past. It turns out that there are pathological cases in which causal past is not infinitely distributive. Thus, if we take infinitely many causal pasts and ask what their common causal past is, the way in which we select the overlap can lead to different outcomes. This is equivalent to saying that modus ponens in causal history logic, is not guaranteed.

1.4 Summarized Causal Logic

Table 1: Correspondence Between Lattice Theory, Causal Structure, and Analogous Logic

Lattice Theory	Notation	Causal Structure Logic	Logic Analogue
Top Element	$\top$	Entire Universe / Lattice (L)	Truth / Tautology ($\top$, True)
Bottom Element	$\perp$	Empty Set ($\emptyset$)	Falsity / Absurdity ($\perp$, False)
Lattice Order	$\leq$	Causal Past Inclusion ($\subseteq$)	Logical Entailment ($\vdash$)
Meet (Infimum)	$\wedge$	Set-Theoretic Intersection ($\cap$)	Conjunction (AND , $\wedge$)
Join (Supremum)	$\vee$	Dedekind Closure over Union ($\overline{\cup}$)	Disjunction (OR , $\vee$)

Additionally, the element $x \implies y$ as seen in Def. 1.19 is the model of implication, but we will see this is not a guaranteed structure in causal logic.

2 IM Lattice Does Not Induce a Heyting Algebra

Remark 2.1. We write $\overline{X}$ to mean the completion of the causal past X . This is the same as $(X^u)^l$. We also write $CP(ct, x, y, z)$ to mean the causal past of the event (ct, x, y, z) .

In order to prove the no-go theorem, we first need to show a lemma. This lemma is a canonical example of *vacuous truth statements*. An example of a vacuous truth: ‘Every pink elephant speaks German.’ Fundamentally this is just the logical rule that false implies true.

This is a proof that the closure of an ever-expanding causal-past is the actual universe itself.

Lemma 2.1. *Given the causal past $B_i = CP(i, i)$, $i \in \mathbb{N}$, we have $\overline{(\bigcup_{i=1}^{\infty} B_i)} = \top$.*

Proof. Suppose there exists an event $X = (t_0, x_0) \in M$ such that for all $n \in \mathbb{N}$, $(n, n) \prec (t_0, x_0)$. By Archimedean Principle, there exists $n_0 \in \mathbb{N}$ such that $t_0 \leq n_0$. Thus, if (n_0, n_0) is a timelike event to X , then $(t_0, x_0) \prec (n_0, n_0)$. Otherwise $(t_0, x_0) \not\prec (n_0, n_0)$. Therefore $(\bigcup_{i=1}^{\infty} B_i)^u = \emptyset$. Now take $\emptyset^l = \{y \in M \mid \forall x \in \emptyset, y \prec x\}$. The condition for the lower set is vacuously true for every $y \in M$. Therefore, $\emptyset^l = \top$. $\square$

We will now prove the no-go theorem. To do this, we use theorem 1.2 that all complete Heyting algebras must be infinitely distributive. We know the IM Lattice is complete under DM completion. Thus, if it is a Heyting algebra, it must also be infinitely distributive. We provide a counterexample using lemma 2.1 to demonstrate a failure of infinite distributivity. This is enough to prove that IM Lattices are not Heyting algebras.

Theorem 2.1 (No-Go on IM Lattice Modus Ponens). *Let L be an IM Lattice. There exists a counter example demonstrating that the IM lattice is not Heyting.*

Proof. We will prove this using a counterexample embedded in 2D Minkowski space⁶. Let $A = CP(1, -1)$ denote a causal past of events before $ct = 1$ and $x = -1$. Let $B_i = CP(i, i)$, $i \in \mathbb{N}$. Then, by lemma 2.1, $A \cap (\bigcup_{i=1}^{\infty} B_i) = A \cap \top = A = CP(1, -1)$. But $A \cap B_i = CP(0, 0)$ for all i (see Fig. 2). Thus $\bigcup_{i=1}^{\infty} (A \cap B_i) = CP(0, 0)$. Take $(0, -1) \in CP(1, -1)$. $(0, -1) \notin CP(0, 0)$. Therefore $CP(1, -1) \neq CP(0, 0)$. $\square$

⁶Keep in mind that the intersection of causal pasts is essentially the overlapping past timelike regions given by the light cones of our causal pasts.

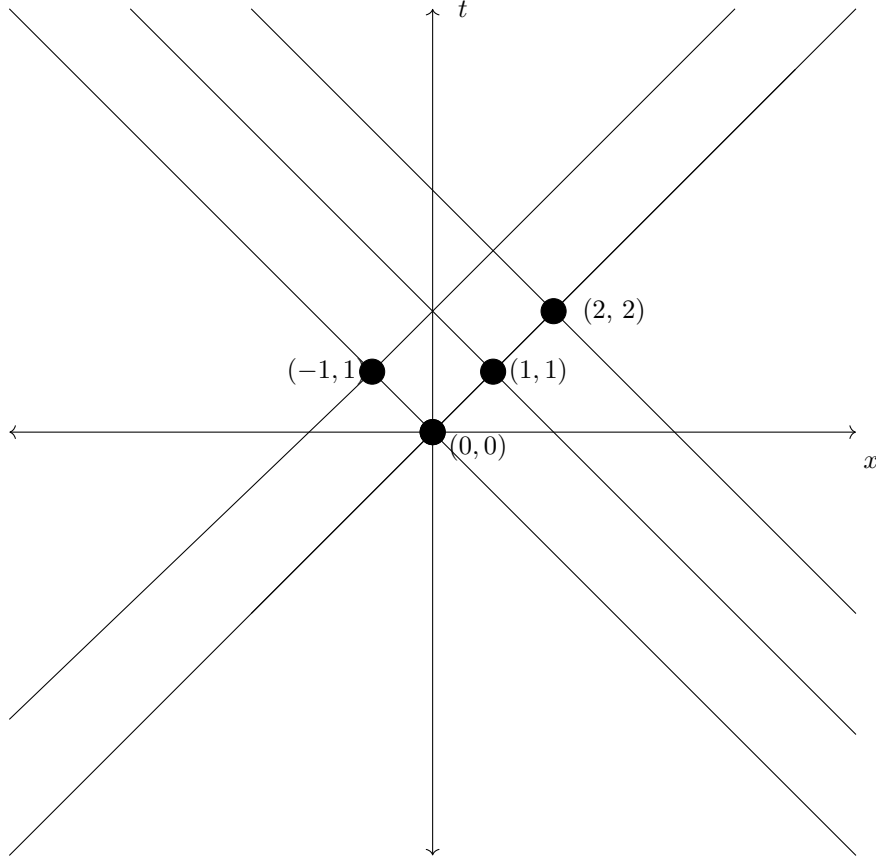


Figure 2: Depiction of overlapping light cones for distributivity $((1, -1) \cap (1, 1)) \cup (1, -1) \cap (2, 2)$.

This gives us a no-go theorem. Essentially we have proved the internal logic of completed causal structure is not powerful enough to guarantee an analogue of *modus ponens* deduction about causal histories. This is reminiscent of von-Neumann’s (infamous) quantum logic, which fails distributivity.

References

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